Suraj Dole

AWS Certified Solutions Architect
Pune, India
+91-7304292526
surajdole007@gmail.com
linkedin.com/in/suraj-dole-b40b8518b

Summary

AWS Certified Solutions Architect with 5+ years of experience in DevOps, cloud infrastructure, and Kubernetes-based microservices. Expertise in designing scalable, highly available, and fault-tolerant AWS architectures using EKS, VPC, and CI/CD pipelines. Strong experience in Infrastructure as Code (Terraform), GitOps (ArgoCD), and containerization (Docker). Proven ability to automate deployments, optimize system performance, and manage production-grade distributed systems. Experienced in AI/ML platform infrastructure, event-driven microservices, and Linux-based system operations across multi-environment Kubernetes clusters.

Skills

Cloud Platforms: AWS (EKS, EC2, VPC, S3, IAM, RDS, API Gateway, CloudWatch, Secrets Manager, Route 53, DocumentDB, ECR)
DevOps & CI/CD: GitLab CI/CD, Jenkins, ArgoCD (GitOps), Argo Image Updater, SonarQube, JFrog Artifactory
Containerization & Orchestration: Docker (multi-stage builds), Kubernetes, Helm, HPA, Taints/Tolerations, Node Affinity
Infrastructure as Code: Terraform (Modules, Workspaces, Remote Backend, multi-env provisioning)
Monitoring & Logging: Prometheus, Grafana, CloudWatch (dashboards, alarms, log insights)
Networking: VPC, Subnets, Route Tables, NAT/IGW, Security Groups, NLB/ALB, Transit Gateway, Site-to-Site VPN
Databases & Messaging: RDS PostgreSQL, DocumentDB, Redis
Security: IAM roles, IRSA, Kubernetes RBAC, SAST scanning, DevSecOps, Secrets Manager

Other: Microservices Architecture, Kafka, RBAC, DevSecOps, Auto-scaling, High Availability, Fault Tolerance, Cost Optimisation, Keycloak, Azure Container Registry (ACR)

Professional Experience

Publicis Sapient May 2025 – Present
Sr. Associate Infrastructure Engineer — Pune, India

- Architected and deployed AWS infrastructure using VPC, EKS, ALB/NLB, and Route 53 ensuring high availability and Fault-Tolerance across multiple availability zones.
- Implemented IAM roles and IRSA for secure, least-privilege access control across Kubernetes workloads, eliminating hardcoded credentials.
- Managed Kubernetes-based AI/ML microservices using Helm, improving scalability and deployment consistency in production environments.
- Implemented CI/CD pipelines using GitLab with automated testing, SAST scanning, and Docker multi-stage image builds with publishing to ECR/ACR.
- Designed GitOps workflows using ArgoCD with Argo Image Updater enabling fully automated, version-controlled deployments across dev/demo/prod environments.
- Integrated backend services with RDS PostgreSQL, DocumentDB, Redis, and S3; configured VPC Links and NLBs for seamless internal service communication.

- Reduced incident resolution time by 40% using Prometheus, Grafana, and CloudWatch monitoring with custom dashboards and alerting pipelines.
- Enforced Kubernetes RBAC by creating Roles, ClusterRoles, RoleBindings, and ClusterRoleBindings to enable least-privilege access for ArgoCD and application service accounts.
- Ensured stable cluster performance by tuning resource requests/limits, taints, tolerations, and node affinity rules; implemented HPA for auto-scaling ML workloads.
- Resolved production-critical issues including CrashLoopBackOff, API Gateway routing failures, CORS errors, and Redis/DocumentDB connectivity problems under SLA.

Tata Technologies Sep 2022 – May 2025

Sr. Solution Developer (Platform/DevOps) — Pune, India

- Developed Connected Vehicle Platform (CVP) handling real-time telemetry data from vehicle TCUs using MQTT and Kafka across 4 Kubernetes clusters.
- Built distributed microservices architecture integrated with Cassandra for distributed data storage and Kafka for high-throughput real-time event processing.
- Implemented event-driven architecture for real-time data processing, ensuring scalability, fault tolerance, and high availability under production load.
- Automated CI/CD pipelines using Jenkins and ArgoCD, reducing deployment effort by 50% and enabling consistent delivery across environments.
- Managed AWS infrastructure (EC2, VPC, NLB, S3) supporting vehicle platform workloads; maintained all services on Linux-based environments.
- Performed code quality enforcement through SonarQube integration with GitLab, ensuring security, maintainability, and test coverage standards.

SLK Automation Dec 2020 – Aug 2022

DevOps Engineer — Bangalore, India

- Implemented CI/CD pipeline automation and Docker-based deployments, improving release frequency and consistency.
- Supported Kubernetes workloads and AWS infrastructure setup, monitoring, and troubleshooting in Linux environments.
- Automated routine infrastructure operations using Bash and Shell scripting, reducing manual effort significantly.
- Configured and maintained infrastructure monitoring tools to ensure system health and proactive alerting.

Projects

Bodhi Platform (AI/ML Microservices Platform)

- Designed and implemented end-to-end AWS architecture for AI/ML workloads using EKS, VPC, ALB/NLB, API Gateway, Route 53, and Secrets Manager.
- Deployed and managed AI/ML microservices (LLM gateway, vision, forecasting, video generation, orchestration) on Kubernetes using Helm charts.
- Built scalable and highly available microservices architecture supporting distributed production workloads across multiple namespaces and clusters.
- Built and optimized GitLab CI pipelines by deploying GitLab Runners on AWS EKS, enabling scalable CI execution for image builds, unit tests with coverage, SAST scans, and publishing Docker images to ECR/ACR.
- Designed GitOps-based continuous delivery using ArgoCD with Argo Image Updater auto-committing updated image tags to Helm manifests, enabling fully automated deployments.
- Integrated backend systems using RDS PostgreSQL, DocumentDB, Redis, and S3 with internal service networking via VPC Links and NLBs.
- Configured secure access using IAM roles, IRSA, and Kubernetes RBAC; managed Secrets Manager for all sensitive configuration.
- Implemented monitoring, logging, and alerting using Prometheus, Grafana, and CloudWatch

covering pod health, API latency, and ML inference performance.
- Optimized Kubernetes workloads using HPA, resource limits, node affinity, taints/tolerations, and horizontal scaling strategies.
- Troubleshot production issues including CrashLoopBackOff, API latency spikes, and database connectivity problems with zero unplanned downtime.

Connected Vehicle Platform (CVP)

- Designed a Connected Vehicle Platform (CVP) to process telemetry data from vehicle TCUs via MQTT protocol, ingested and distributed across 4 Kubernetes clusters.
- Managed microservices architecture integrated with Cassandra for distributed data storage and Kafka for real-time telemetry event processing.
- Implemented AWS cloud infrastructure using EC2, VPC, NLB, and S3; automated deployments via Jenkins CI/CD pipelines and ArgoCD for GitOps-based delivery.
- Built and maintained Docker containerization for all services; ensured reliability and scalability of the platform under high-frequency vehicle data ingestion.

Education

B.Tech in Electronics and Telecommunications Engineering
Rajarambapu Institute of Technology, Sangli 2016 – 2020

Certification

AWS Certified Solutions Architect – Associate